[457] On the Quartic Surfaces $(*\,|\,U, V, W)^2 = 0$.

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I propose to myself for investigation the quartic surfaces represented by an equation

$$(*\,|\,U,\,V,\,W)^2 = 0,$$

where U, V, W are quadric functions of the coordinates.

Such a surface has 8 nodes (conical points), viz., these are the points of intersection of the quadric surfaces $U = 0$, $V = 0$, $W = 0$. It is to be observed, that not every quartic surface with 8 nodes is included under the above form; in fact the equation of a quartic surface contains (homogeneously) 35 coefficients, or say 34 arbitrary parameters; in order that a given point may be a node, 4 conditions must be satisfied, and it is consequently possible to find a quartic surface having 8 given points as nodes (and having in its equation $(34 - 8 \cdot 4 =)$ 2 arbitrary parameters): but 8 given points are not in general the intersections of three quadric surfaces, and such a quartic surface is therefore not in general included under the above form. I think, however, that it may be assumed that the above form includes all the quartic surfaces having 8 nodes, points of intersection of three quadric surfaces. It will presently appear that, included in the form, we have surfaces where (instead of the 8 nodes) there is a nodal or cuspidal conic; and that these are the most general forms of such quartic surfaces.

A quartic surface has at most 16 nodes, and the general form with 8 nodes must admit of being particularised so that the surface shall acquire any number not exceeding 8 of additional nodes. This does not show, but it is probable, that the above special form with 8 nodes can be particularised so that the surface shall in like manner acquire any number not exceeding 8 of additional nodes. Similarly, a quartic surface with a nodal conic may have besides 1, 2, 3, or 4 nodes; and it will be shown in the sequel how the form, particularised so as to give a nodal conic, may be further particularised so as to give the 1, 2, 3, or 4 nodes. So a quartic surface with a cuspidal conic may besides have 1 node, and it will be shown how the form, particularised so as to give a cuspidal conic, may be further particularised so as to give 1 node.

Starting from the equation $(*\,|\,U, V, W)^2 = 0$, we may, by substituting for U, V, W linear functions of these expressions, transform the equation precisely in the manner of a conic, and therefore into any of the forms under which the equation of a conic can be exhibited; for instance, in the forms

$$aU^2 + bV^2 + cW^2 = 0, \qquad fVW + gWU + hUV = 0, \qquad UW - V^2 = 0, \quad \&c.$$

I attend at present only to the last-mentioned form $UW - V^2 = 0$, which, it thus appears, is equally general with the original form $(*\,|\,U, V, W)^2 = 0$.

The quartic surface

$$UW - V^2 = 0,$$

where U, V, W are any quadric functions of the coordinates, may be considered as the envelope of the quadric surface

$$(U, V, W\,|\,\theta, 1)^2 = 0,$$

where θ is an arbitrary parameter. And it thus appears that it is very easy to reciprocate (in regard to any given quadric surface) the quartic surface. For the reciprocal of the quartic surface is clearly the envelope of the reciprocal of the variable quadric surface; this reciprocal is itself a quadric surface, and the reciprocal of the quartic surface is thus given in the same form as the original surface, viz., as the envelope of a quadric surface the equation whereof contains rationally the variable parameter θ; the equation of the reciprocal surface is consequently obtained by equating to zero the discriminant in regard to θ, of the equation of the reciprocal quadric surface.

It is to be observed that, inasmuch as the equation of the reciprocal quadric surface is of the third degree in the coefficients of the original quadric, it is in general of the degree 6 in the parameter; we have thus a sextic function of θ, the coefficients whereof are quadric functions of the coordinates; and the discriminant is a function of the order 10 in these coefficients, that is, of the order 20 in the coordinates. The reciprocal of the quartic surface is thus a surface of the order 20; this is right, for in a general quartic surface the order of the reciprocal surface is 36, and the 8 nodes reduce the order by 16; $36 - 16 = 20$.

In the equation $UW - V^2 = 0$, if U reduce itself to the square of a linear function, $= P^2$, the equation becomes $V^2 - P^2W = 0$, which is the general form of the quartic surface having the nodal conic $V = 0$, $P = 0$. And if, moreover, W be the product of this same linear function P by another linear function Q, $W = PQ$, then the form is $V^2 - P^3Q = 0$, which is the general form of the quartic surface having the cuspidal conic $V = 0$, $P = 0$.

Writing for greater convenience x, y in the place of P, Q respectively, we have the quartic

$$V^2 - x^3y = 0, \tag{AA}$$

having the cuspidal conic $V = 0$, $x = 0$; and which has besides the conic of plane contact $V = 0$, $y = 0$. In virtue of the cuspidal conic the reciprocal surface should be of the order 6; and by the foregoing method of obtaining the equation of the reciprocal surface, I will verify that this is so. To effect this as simply as possible, I fix the remaining coordinates z, w as follows. The line $x = 0$, $y = 0$ is not in general a tangent to the surface $V = 0$; it therefore meets this surface in two points, and we may take $z = 0$, $w = 0$ to be the equations of the tangent planes at these two points respectively; we have thus $V = ax^2 + 2hxy + by^2 + 2nzw$. Introducing for convenience the numerical factor 2, and taking the equation of the surface to be

$$(ax^2 + 2hxy + by^2 + 2nzw)^2 - 2x^3y = 0,$$

this is the envelope of the quadric surface

$$\theta^2 \cdot x^2 + 2\theta(ax^2 + 2hxy + by^2 + 2nzw) + 2xy = 0,$$

which is a surface $(\mathfrak{a}, \mathfrak{b}, \mathfrak{c}, \mathfrak{d}, \mathfrak{f}, \mathfrak{g}, \mathfrak{h}, \mathfrak{l}, \mathfrak{m}, \mathfrak{n} \,|\, x, y, z, w)^2 = 0$, where $\mathfrak{a} = \theta^2 + 2a\theta$, $\mathfrak{b} = 2b\theta$, $\mathfrak{h} = 2\theta h + 1$, $\mathfrak{n} = 2\theta n$, and where all the other coefficients vanish. Assuming, as usual, that the reciprocation is effected in regard to the surface $x^2 + y^2 + z^2 + w^2 = 0$, the general equation is

$$\begin{aligned}
&x^2 \cdot \mathfrak{d}(\mathfrak{bc} - \mathfrak{f}^2) - \mathfrak{cm}^2 - \mathfrak{bn}^2 + 2\mathfrak{fmn} \\
&+ y^2 \cdot \mathfrak{d}(\mathfrak{ca} - \mathfrak{g}^2) - \mathfrak{an}^2 - \mathfrak{cl}^2 + 2\mathfrak{gnl} \\
&+ z^2 \cdot \mathfrak{d}(\mathfrak{ab} - \mathfrak{h}^2) - \mathfrak{bl}^2 - \mathfrak{am}^2 + 2\mathfrak{hlm} \\
&+ w^2 \cdot \mathfrak{abc} - \mathfrak{af}^2 - \mathfrak{bg}^2 - \mathfrak{ch}^2 + 2\mathfrak{fgh} \\
&+ 2yz \cdot \mathfrak{d}(\mathfrak{gh} - \mathfrak{af}) + \mathfrak{l}^2\mathfrak{f} + \mathfrak{amn} - \mathfrak{hnl} - \mathfrak{glm} \\
&+ 2zx \cdot \mathfrak{d}(\mathfrak{hf} - \mathfrak{bg}) + \mathfrak{m}^2\mathfrak{g} + \mathfrak{bnl} - \mathfrak{fhn} - \mathfrak{hmn} \\
&+ 2xy \cdot \mathfrak{d}(\mathfrak{fg} - \mathfrak{ch}) + \mathfrak{n}^2\mathfrak{h} + \mathfrak{clm} - \mathfrak{gmn} - \mathfrak{fnl} \\
&+ 2xw \cdot -\mathfrak{l}(\mathfrak{bc} - \mathfrak{f}^2) - \mathfrak{m}(\mathfrak{hf} - \mathfrak{bg}) - \mathfrak{n}(\mathfrak{fg} - \mathfrak{ch}) \\
&+ 2yw \cdot -\mathfrak{m}(\mathfrak{ca} - \mathfrak{g}^2) - \mathfrak{l}(\mathfrak{fg} - \mathfrak{ch}) - \mathfrak{n}(\mathfrak{gh} - \mathfrak{af}) \\
&+ 2zw \cdot -\mathfrak{n}(\mathfrak{ab} - \mathfrak{h}^2) - \mathfrak{m}(\mathfrak{gh} - \mathfrak{af}) - \mathfrak{l}(\mathfrak{hf} - \mathfrak{bg}),
\end{aligned}$$

(I write down this general result as it will be useful for reference in other cases); in the present case this becomes simply

$$x^2 \cdot b\mathfrak{n}^2 + y^2 \cdot a\mathfrak{n}^2 + 2xy \cdot \mathfrak{n}\mathfrak{h}^2 + 2zw(-\mathfrak{n}ab + \mathfrak{n}\mathfrak{h}^2) = 0,$$

where a, b, $\mathfrak{n}$, $\mathfrak{h}$ have the foregoing values; the equation is thus only of the order 4 in regard to θ; but it in fact divides by $\mathfrak{n}\,(= 2\theta n)$ and thus reduces itself to the third order, viz. it becomes

$$\mathfrak{n}(bx^2 + ay^2) - 2\mathfrak{h}^2 xy + 2(ab - \mathfrak{h}^2)zw = 0,$$

or, substituting for $\mathfrak{a}$, $\mathfrak{h}$, $\mathfrak{n}$, $\mathfrak{b}$ their values, this is

$$x^2 \cdot 4bn\theta^3 + 2y^2(2n\theta^3 + 4an\theta^2) + 2zw(2b\theta^2 + 4ab\theta) - 2(xy + zw)(2\theta h + 1)^2 = 0,$$

say this is

$$(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{D}\,|\,\theta, 1)^3 = 0,$$

where $\mathfrak{A}$, $\mathfrak{B}$, $\mathfrak{C}$, $\mathfrak{D}$ are each of them a quadric function of the coordinates; it being observed that $\mathfrak{C}$ and $\mathfrak{D}$ are respectively numerical multiples of the same function $(xy + zw)$. Hence equating the discriminant to zero, we have

$$\mathfrak{A}^2\mathfrak{D}^2 + 4\mathfrak{A}\mathfrak{C}^3 + 4\mathfrak{B}^3\mathfrak{D} - 3\mathfrak{B}^2\mathfrak{C}^2 - 6\mathfrak{A}\mathfrak{B}\mathfrak{C}\mathfrak{D} = 0,$$

which equation, inasmuch as every term contains either $\mathfrak{C}$ or $\mathfrak{D}$ as a factor, divides by $xy + zw$, and thus becomes an equation of the order 6 in the coordinates: that is, the order of the reciprocal surface is 6. Multiplying by $\frac{1}{8}$ to avoid fractions, the actual values of $\mathfrak{A}$, $\mathfrak{B}$, $\mathfrak{C}$, $\mathfrak{D}$ are

$$\begin{aligned}
\mathfrak{A} &= b\,n^2 + 3(ny^2 + 2hzw),\\
\mathfrak{B} &= 2\{hnx^2 + any^2 + 2ahzw - 2h^2(xy + zw)\},\\
\mathfrak{C} &= -4fh\,(xy + zw),\\
\mathfrak{D} &= -3\,(xy + zw),
\end{aligned}$$

or say $\mathfrak{A} = 3\alpha$, $\mathfrak{B} = 2\beta$, $\mathfrak{C} = -4h\gamma$, $\mathfrak{D} = -3\gamma$; where $\gamma = xy + zw$; substituting these values and omitting the factor 8γ, the equation is

$$27\alpha^3\gamma - 256h^4\alpha\gamma^3 - 32\beta^3 - 64h^3\beta^2\gamma^2 - 144h\alpha\beta^2\gamma = 0,$$

which is an equation of the form $(*\,|\,\alpha, \beta, \gamma)^3 = 0$. The sextic surface has thus singular points $\alpha = 0$, $\beta = 0$, $\gamma = 0$, viz. these are the two points $(x = 0, y = 0, z = 0)$, $(x = 0, y = 0, w = 0)$ each four times. The further discussion of the sextic surface is reserved for another occasion.

I do not at present attempt to enumerate the particular cases of the surface $V^2 - x^3y = 0$, but content myself with the discussion of a particular case in which the order of the reciprocal surface is $= 3$. Suppose that $V = 0$ is a cone, $y = 0$ a tangent plane to the cone (so that the conic $y = 0$, $V = 0$ breaks up into a line twice repeated), $x = 0$ an arbitrary plane (so that we have still the proper cuspidal conic $x = 0$, $V = 0$). Any other tangent plane of the cone may be taken for the plane $z = 0$; the plane containing the lines of contact of the two tangent planes for the plane $w = 0$; the equation of the conic then is $V = dw^2 + 2fyz = 0$; and the equation of the surface is

$$(dw^2 + 2fyz)^2 - x^3y = 0. \tag{AB}$$

For convenience of comparison, I change x, y, w, z into y, w, z, x, and assign numerical values to the coefficients, writing the equation under the form

$$27(4aw + z^2)^2 - 64y^3w = 0. \tag{AB}$$

The quartic is here the envelope of the quadric surface

$$\theta^2 \cdot 4yw + \theta \cdot 9(4aw + z^2) + 12yz = 0,$$

viz., comparing with the general form

$$0 = (\mathfrak{a}, \mathfrak{b}, \mathfrak{c}, \mathfrak{d}, \mathfrak{f}, \mathfrak{g}, \mathfrak{h}, \mathfrak{l}, \mathfrak{m}, \mathfrak{n} \,|\, x, y, z, w)^2,$$

we have $\mathfrak{b} = 12$, $\mathfrak{c} = 9\theta$, $\mathfrak{l} = 18\theta$, $\mathfrak{m} = 2\theta^2$, all the other coefficients vanishing. The reciprocal equation is

$$\mathfrak{d}\mathfrak{m}^2 + x^2(-\mathfrak{c}\mathfrak{m}^2) + y^2(-\mathfrak{c}\mathfrak{l}^2) + z^2(-\mathfrak{b}\mathfrak{l}^2) + 2xw \cdot \mathfrak{m}^2\mathfrak{g} + 2xw(-\mathfrak{l}\mathfrak{b}\mathfrak{c}) = 0,$$

or substituting for $\mathfrak{b}$, $\mathfrak{c}$, $\mathfrak{l}$, $\mathfrak{m}$ their values, this is found to be

$$\theta(\theta x - 9y)^2 + 108(z^2 + xw) = 0.$$

Representing this by $(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{D} \,|\, \theta, 1)^3 = 0$, the discriminant in regard to θ would, in virtue of the values of $\mathfrak{A}$, $\mathfrak{B}$, $\mathfrak{C}$, contain $\mathfrak{D}$ as a factor; the reason of this appears from the original form; in fact, forming the derived equation in regard to θ, this is found to be $(\theta x - 9y)(\theta x - 3y) = 0$; the value $\theta x - 9y = 0$ gives as a factor of the discriminant $z^2 + xw$; the value $\theta x - 3y = 0$ gives $3y(-6y)^2 + 108x(z^2 + xw)$, that is the factor $y^3 + x(z^2 + xw)$; the complete value of the discriminant as obtained by substitution of the values of $\mathfrak{A}$, $\mathfrak{B}$, $\mathfrak{C}$, $\mathfrak{D}$ being $x^2(z^2 + xw)\{y^3 + x(z^2 + xw)\}$; the equation of the reciprocal surface is

$$y^3 + x(z^2 + xw) = 0,$$

viz. this is a cubic surface. Prof. Schläfli's Case XX., having a uniplanar point $x = 0$, $y = 0$, $z = 0$ reducing the class by 8, and so giving a reciprocal surface of the order $(12 - 8 =)\, 4$, viz. the surface $27(4xw + z^2)^2 - 64y^3w = 0$. See the Memoir, Schläfli, "On the distribution of surfaces of the third order into species in reference to the absence or presence of singular points and the reality of their lines," *Phil. Trans.*, vol. CLIII. (1853), pp. 193–241.

I pass to the case of a surface

$$V^2 - P^2U = 0$$

having a nodal conic $V = 0$, $P = 0$, but not having in general any nodes. And I propose to show how the constants may be determined so that the surface shall have 1, 2, 3, or 4 nodes. It is to be remarked that in the above equation the plane $P = 0$ is a determinate plane, but the quadric surface $U = 0$ is not a determinate quadric, we may in fact substitute for it the quadric $U + \lambda P^2 = 0$, writing the equation under the form

$$V^2 + 2\lambda V P^2 + \lambda^2 P^4 = P^2(U + \lambda^2 P^2),$$

so that we may without loss of generality, by means of the disposable constant λ, subject the surface $V = 0$ to any single condition; for instance, take it to be a cone, or to pass through a given point, &c.

Taking the planes $x = 0$, $y = 0$, $z = 0$, $w = 0$ to be arbitrary planes, the implicit constant factors in these equations may be determined in such wise that the equation of the given plane $P = 0$ shall be $x + y + z + w = 0$. The equation of the surface will then be

$$\begin{aligned}&\{(a, b, c, d, f, g, h, l, m, n \,|\, x, y, z, w)^2\}^2 \\ &\qquad = (x + y + z + w)^2 \cdot (a', b', c', d', f', g', h', l', m', n' \,|\, x, y, z, w)^2,\end{aligned}$$

and we may assume that the node or nodes (if any) lie at a vertex or vertices of the tetrahedron $x = 0$, $y = 0$, $z = 0$, $w = 0$, say at the points A, B, C, D. The conditions for a node at each of these points are at once found to be

Node at A,	Node at B,	Node at C,	Node at D,
$2a^2 = a'$,	$2bh = h' + b'$,	$2cg = g' + c'$,	$2dl = l' + d'$,
$2ah = h' + a'$,	$2b^2 = b'$,	$2cf = f' + c'$,	$2dm = m' + d'$,
$2ag = g' + a'$,	$2hf = f' + h'$,	$2c^2 = c'$,	$2dn = n' + d'$,
$2al = l' + a'$,	$2bm = m' + b'$,	$2cn = n' + c'$,	$2d^2 = d'$.

The first set of equations gives

$$a' = a^2, \quad h' = 2ah - a^2, \quad g' = 2ag - g^2, \quad l' = 2al - a^2.$$

If the first and second sets are satisfied simultaneously, we have $2(a-b)h = a^2 - b^2$, that is $a = b$, or else $h = \frac{1}{2}(a+b)$; that is, the two sets may be satisfied in two different ways according as a and b are equal or unequal. Similarly the first, second, and third sets may be satisfied in three different ways and the four sets in five different ways according as there are or are not any equalities between a, b, c, and between a, b, c and d respectively. The several solutions are shown in the annexed table, viz., in the line I no set is satisfied; in the line II only the first set; in the lines III and IV the first and second sets; in the lines V to VII the first, second, and third sets; and in the lines VIII to XII the four sets.

[See next page for this Table, which should come in here.]

I is the general case, $V^2 = P^2U$, of a quartic and a nodal conic but without nodes. (AC).

II is the case of a single node; writing, as without loss of generality we may do, $a = 0$, the equation is

$$\{(0, b, c, d, f, g, h, l, m, n \,|\, x, y, z, w)^2\}^2 = (x+y+z+w)^2 \cdot (b, c, d, n, m, f \,|\, y, z, w)^2,$$

viz., the quadric $U = 0$ is here a cone having its vertex on the quadric $V = 0$. (AD).

III and IV are two cases each of them with two nodes, viz. III, the equation is

$$\begin{aligned}\{(ax+by)(x+y) + cz^2 + 2nzw + dw^2 + 2x(gz+lw) + 2y(fz+mw)\}^2 \\ = (x+y+z+w)^2\,[(ax+by)^2 + c'z^2 + 2n'zw + d'w^2 \\ + 2x\{(2ag-a^2)z + (2al-a^2)w\} + 2y\{(2bf-b^2)z + (2bm-b^2)w\}], \quad (AE)\end{aligned}$$

where it is to be observed that the line $z = 0$, $w = 0$ joining the two nodes $(y = 0, z = 0, w = 0)$ and $(w = 0, z = 0, x = 0)$ is a line on the surface. Writing, as we may do, $a = 0$, the equation assumes the more simple form

$$\begin{aligned}\{by(x+y) + cz^2 + 2nzw + dw^2 + 2x(gz+lw) + 2y(fz+mw)\}^2 \\ = (x+y+z+w)^2\,[b^2y^2 + c'z^2 + 2n'zw + d'w^2 + 2y\{(2bf-b^2)z + (2bm-b^2)w\}]. \quad (AE)\end{aligned}$$

[Page 310 of the original carries here a full-page sideways table classifying the twelve cases I–XII according to the equalities between a, b, c, d and the consequent values of a', b', c', d', h', g', l', f', m', n'. The table is reproduced in schematic form below.]

Line	Conditions on a, b, c, d	Number of nodes
I	all distinct, no set satisfied	0
II	1st set only; $a = 0$	1
III	1st and 2nd sets; a, b distinct	2
IV	1st and 2nd sets; $a = b$	2
V	1st, 2nd, 3rd sets; a, b, c all distinct	3
VI	1st, 2nd, 3rd sets; one equality	3
VII	1st, 2nd, 3rd sets; $a = b = c$	3
VIII	all four sets; a, b, c, d all distinct	4
IX	all four sets; one pair equal	4
X	all four sets; two pairs equal	4
XI	all four sets; three equal	4
XII	all four sets; $a = b = c = d$	4

In IV the equation is

$$\begin{aligned}&\{a(x^2+y^2)+2hxy+cz^2+dw^2+2nzw+2x(gz+lw)+2y(fz+mw)\}^2\\ &\qquad=(x+y+z+w)^2\,[a^2(x^2+y^2)+2(2ah-a^2)xy+c'z^2+2n'zw+d'w^2\\ &\qquad\qquad+2x\{(2ag-a^2)z+(2al-a^2)w\}+2y\{(2af-a^2)z+(2am-a^2)w\}],\quad (AF)\end{aligned}$$

where it will be observed that the line $z=0$, $w=0$ joining the two nodes is not a line on the surface.

Writing, as we may do, $a=0$, the equation becomes

$$\begin{aligned}&\{2hxy+cz^2+2nzw+dw^2+2x(gz+lw)+2y(fz+mw)\}^2\\ &\qquad=(x+y+z+w)^2(c'z^2+2n'zw+d'w^2),\quad (AF)\end{aligned}$$

viz. the form is $V^2=P^2QR$, the quadric surface $U=0$ breaking up into the two planes $Q=0$, $R=0$; and the nodes being situate at the intersections of the line $Q=0$, $R=0$ with the surface $V=0$.

V, VI, VII are apparently cases with three nodes, but in fact VI is the only case of a proper quartic surface with three nodes. For in V the equation is

$$\begin{aligned}&\{(ax+by+cz)(x+y+z)+dw^2+2w(lx+my+nz)\}^2\\ &\qquad=(x+y+z+w)^2\,[(ax+by+cz)^2+d'w^2\\ &\qquad\qquad+2w\{(2al-a^2)x+(2bm-b^2)y+(2cn-c^2)z\}],\end{aligned}$$

which is satisfied by $w=0$, and the surface thus breaks up into the plane $w=0$ and a cubic surface.

And in VII the equation is

$$\begin{aligned}&\{a(x^2+y^2+z^2)+dw^2+2fyz+2gzx+2hxy+2lxw+2myw+2nzw\}^2\\ &\qquad=(x+y+z+w)^2\,[a^2(x^2+y^2+z^2)+d^2w^2\\ &\qquad+2\{(2af-a^2)yz+(2ag-a^2)zx+(2ah-a^2)xy\\ &\qquad\qquad+(2al-a^2)xw+(2am-a^2)yw+(2an-a^2)zw\}],\end{aligned}$$

which putting $a=0$ is

$$(dw^2+2fyz+2gzx+2hxy+2lxw+2myw+2nzw)^2=(x+y+z+w)^2\,d'^2w^2,$$

viz., this is a pair of quadric surfaces.

In the remaining case VI the equation is

$$\begin{aligned}&\{a(x^2+y^2)+2hxy+cz^2+dw^2+(a+c)(yz+zx)+2w(lx+my+nz)\}^2\\ &\qquad=(x+y+z+w)^2\,[a^2(x^2+y^2)+c'^2z^2+d'w^2+2acz(x+y)+2(2ah-a^2)xy\\ &\qquad\qquad+2w\{(2al-a^2)x+(2am-a^2)y+(2cn-c^2)z\}],\quad (AG)\end{aligned}$$

which putting therein $a=0$ is

$$\begin{aligned}&\{2hxy+dw^2+cz(x+y+z)+2w(lx+my+nz)\}^2\\ &\qquad=(x+y+z+w)^2\,\{c'^2z^2+d'w^2+2(2cn-c^2)zw\},\quad (AG)\end{aligned}$$

which is a surface having the nodes A, B, C; and it is to be observed that the lines CA, CB, but not the line AB, are lines on the surface.

IX, X, XI, XII are apparently cases with four nodes, but it is only XI which is a proper quartic with four nodes. In fact IX is

$$\{(ax+ay+cz+dw)(x+y+z+w)+2(h-a)xy\}^2 = (x+y+z+w)^2\{(ax+ay+cz+dw)^2+4a(h-a)xy\},$$

which is satisfied if $x = 0$ or if $y = 0$; that is, the surface breaks up into the two planes $x = 0$, $y = 0$, and a quadric surface.

X is

$$\begin{aligned}\{a(x^2+y^2+z^2)+dw^2+2fyz+2gzx+2hxy+(a+d)(xw+yw+zw)\}^2 \\ = (x+y+z+w)^2\{a^2(x^2+y^2+z^2)+d^2w^2 \\ +2(2af-a^2)yz+2(2ag-a^2)zx+2(2ah-a^2)xy+2ad(xw+yw+zw)\},\end{aligned}$$

which putting therein $a = 0$ is

$$\{dw(x+y+z+w)+2fyz+2gzx+2hxy\}^2 = (x+y+z+w)^2\, d^2w^2,$$

and thus breaks up into two quadrics.

And XII is

$$\begin{aligned}\{a^2(x^2+y^2+z^2+w^2)+2fyz+2gzx+2hxy+2lxw+2myw+2nzw\}^2 \\ = (x+y+z+w)^2\,[a^2(x^2+y^2+z^2+w^2) \\ +2(2af-a^2)yz+2(2ag-a^2)zx+2(2ah-a^2)xy \\ +2(2al-a^2)xw+2(2am-a^2)yw+2(2an-a^2)zw\},\end{aligned}$$

which putting $a = 0$ is

$$(2fyz+2gzx+2hxy+2lxw+2myw+2nzw)^2 = 0,$$

and is thus a quadric surface twice repeated.

There remains XI, and here the equation is

$$\begin{aligned}\{(ax+ay+cz+cw)(x+y+z+w)+2(h-a)xy+2(n-c)zw\}^2 \\ = (x+y+z+w)^2\{(ax+ay+cz+cw)^2+4a(h-a)xy+4c(n-c)zw\},\end{aligned}$$

or writing $h+a$, $n+c$ in place of a, c respectively, this is

$$\begin{aligned}\{(ax+ay+cz+cw)(x+y+z+w)+2hxy+2nzw\}^2 \\ = (x+y+z+w)^2\{(ax+ay+cz+cw)^2+4ahxy+4cnzw\}, \quad (AH)\end{aligned}$$

or putting herein $a = 0$ it is

$$\{c(z+w)(x+y+z+w)+2hxy+2nzw\}^2 = c(x+y+z+w)^2\{c(z+w)^2+4nzw\}, \quad (AH)$$

which may also be written

$$c(x+y+z+w)\{hxy(z+w)-nzw(x+y)\}+(hxy+nzw)^2 = 0, \qquad (AH)$$

the equation of a quartic surface with the four nodes A, B, C, D; it is to be observed that the lines AC, AD, BC, BD are, the lines AB and CD are not, lines on the surface.

A more simple form may be given to the equation as follows; using the second of the above forms, multiplying the equation by 4, and writing therein

$$\begin{aligned} p &= \sqrt{c}\,(x+y+z+w), \\ qr &= c(z+w)^2 + 4nzw, \\ st &= c(x+y)^2 - 4hxy, \end{aligned}$$

q, r, and s, t being the linear factors of the two quadric functions respectively, we have

$$qr - st = c(x+y+z+w)(-x-y+z+w) + 4hxy + 4nzw,$$

and thence

$$p^2 + qr - st = 2c(z+w)(x+y+z+w) + 4hxy + 4nzw,$$

wherefore the equation is

$$\tfrac{1}{4}(p^2+qr-st)^2 = st \cdot \tfrac{1}{4}\,p^2 - qr, \qquad (AH)$$

or, what is the same thing,

$$p + \sqrt{qr} + \sqrt{st} = 0, \qquad (AH)$$

where p, q, r, s, t are any linear functions of the coordinates; this is the equation of a quartic surface having the nodal conic $p=0$, $qr-st=0$; and the four nodes $(q=0, r=0, p-\sqrt{st}=0)$ and $(s=0, t=0, p-\sqrt{qr}=0)$. It includes the Cyclide, the equation of which may be written

$$\theta = \sqrt{(ax-ek)^2 + b^2z^2} + \sqrt{(ex-ak)^2 - b^2z^2}.$$

I remark that Prof. Kummer in his most valuable Memoir, "Ueber die Flächen vierten Grades auf welchen Schaaren von Kegelschnitten liegen," *Crelle*, t. LXVI. (1864), pp. 66–76, has considered several of the cases of a quartic surface with a nodal conic, viz. no node, (AC); a single node, (AD); two nodes (the case AF); and four nodes, (AH); but he has not considered two nodes, the case (AE); nor three nodes, (AG).

In reference to the general case of a quartic surface with a nodal conic, some most interesting properties have recently been obtained by Prof. Clebsch, see *Berl. Monatsh.*, April 30, 1868, where it is shown that there are on the surface 16 right lines forming 20 systems of double-fours, analogous in some respect to the 27 lines and 36 systems of double-sixes of a cubic surface.

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[458] On the Anharmonic-Ratio Sextic.

[From the Quarterly Journal of Pure and Applied Mathematics, vol. X. (1870), pp. 56, 57.]

Mr Walker's equation is

$$4A\left(\lambda^2+\lambda+1\right)^3 - 27I^2\left(\lambda^2-\lambda\right)^2 = 0;$$

changing the sign of λ, and also the numerical multipliers of I, A (so as to convert the discriminant equation into its standard form $\Delta = I^3 - 27J^2$), the equation is

$$4\Delta\left(\lambda^2+\lambda+1\right)^3 - 27J^2\left(\lambda^2+\lambda\right)^2 = 0.$$

I remark that this is most readily obtained as follows; writing

$$\begin{aligned} A &= (a-d)(b-c), \\ B &= (b-d)(c-a), \\ C &= (c-d)(a-b), \end{aligned}$$

then we have $A+B+C=0$,

$$\begin{aligned} I &= \tfrac{1}{6}(A^2+B^2+C^2) = -\tfrac{1}{3}(BC+CA+AB),\\ J &= \tfrac{1}{216}(B-C)(C-A)(A-B),\\ \sqrt{\Delta} &= \tfrac{1}{12}\,ABC, \end{aligned}$$

see my Fifth Memoir on Quantics, *Phil. Trans.*, vol. CXLVIII. (1858), pp. 429–460, [156].

And observe also, that in virtue of the relation $A+B+C=0$, we have

$$12I = A^2+AB+B^2 = A^2+AC+C^2 = B^2+BC+C^2.$$

Hence writing

$$u = \frac{A}{B} + \frac{1}{4}\frac{B}{A} + \frac{1}{4},$$

when λ has any one of the values $\frac{A}{B}$, $\frac{B}{A}$, $\frac{A}{C}$, $\frac{C}{A}$, $\frac{B}{C}$, $\frac{C}{B}$, we see that u assumes only the values A, B, C, and u is thus determined by the equation

$$u^2 - 12Iu - 16\sqrt{\Delta} = 0.$$

Eliminating u, we obtain

$$16\sqrt{\Delta}\left\{\frac{4\Delta}{27J^2}\left(\lambda+\tfrac{1}{4}\cdot\tfrac{1}{\lambda}+1\right)^2 - \left(\lambda+\tfrac{1}{4}\cdot\tfrac{1}{\lambda}+1\right) - 1\right\} = 0,$$

or, what is the same thing,

$$4\Delta\left(\lambda^2+\lambda+1\right)^3 - 27J^2\lambda^2\left(\lambda+1\right)^2 = 0,$$

that is

$$4\Delta\left(\lambda^2+\lambda+1\right)^3 - 27J^2\left(\lambda^2+\lambda\right)^2 = 0,$$

the required equation.

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[459] On the Double-sixers of a Cubic Surface.

[From the Quarterly Journal of Pure and Applied Mathematics, vol. X. (1870), pp. 58–71.]

The 27 lines on a cubic surface include, and that in 36 different ways, a double-sixer; viz. a system of two sets of six lines $1, 2, 3, 4, 5, 6$; $1', 2', 3', 4', 5', 6'$, such that every line of the one set intersects all the non-corresponding lines of the other set, thus

	1	2	3	4	5	6
$1'$	·	●	●	●	●	●
$2'$	●	·	●	●	●	●
$3'$	●	●	·	●	●	●
$4'$	●	●	●	·	●	●
$5'$	●	●	●	●	·	●
$6'$	●	●	●	●	●	·

there being in all 30 intersections.

Any line say 4, of the one set, intersects five lines $1', 2', 3', 5', 6'$ of the other set; and these six lines being given the double-sixer may be constructed; viz. (besides the line 4) we have a line 1 meeting the lines $2', 3', 5', 6'$; a line 2 meeting the lines $3', 5', 6', 1'$; a line 3 meeting the lines

$5', 6', 1', 2'$; a line 5 meeting the lines $6', 1', 2', 3'$; and a line 6 meeting the lines $1', 2', 3', 5'$; and then the lines $1, 2, 3, 5, 6$ are all of them met by a single line $4'$, which completes the system.

We may, if we please, consider the lines 4, 2 as given, and then $1', 3', 5', 6'$ will be any four lines each of them meeting the two given lines 4, 2; $2'$ will be any line meeting 4; and we have to determine a line $4'$ meeting 2, such that there may exist the lines $1, 3, 5, 6$, completing the system as above. Or what is the same thing, we have a skew quadrilateral $1', 2, 3', 4$; $5'$ and $6'$ meet 2 and 4; $2'$ meets 4, and $4'$ meets 2: 5 and 6 meet $1'$ and $3'$; 1 meets $3'$ and 3 meets $1'$; and the two sets $2', 4', 5', 6'$ and $1, 3, 5, 6$ meet thus

	1	3	5	6
$2'$	·	·	●	●
$4'$	·	·	●	●
$5'$	●	●	·	·
$6'$	●	●	·	·

Hence, starting with the skew quadrilateral $1'23'4$, and taking $x = 0$, $y = 0$, $z = 0$, $w = 0$ for the equations of the four planes $41'$, $1'2$, $23'$, $3'4$ respectively; or what is the same thing $x = 0$, $y = 0$ for the equations of the line $1'$; $y = 0$, $z = 0$ for those of the line 2; $z = 0$, $w = 0$ for those of the line $3'$; and $w = 0$, $x = 0$ for those of the line 4; the several lines may be determined, each of them by means of its six coordinates, as follows:

	a	b	c	f	g	h
$1'$	·	·	·	·	·	1
2	·	·	·	1	·	·
$3'$	·	·	·	·	1	·
4	1	·	·	·	·	·
$2'$	A_1	B_1	C_1	·	G_1	H_1
$4'$	·	B_2	C_2	F_2	G_2	H_2
$5'$	·	B_3	C_3	F_3	G_3	H_3
$6'$	·	B_4	C_4	F_4	G_4	H_4
1	a_1	b_1	c_1	f_1	g_1	·
3	·	·	·	f_2	g_2	·
5	·	·	·	f_3	g_3	·
6	a_4	b_4	·	f_4	g_4	·

where

$$\begin{aligned} B_1G_1 + C_1H_1 &= 0, \\ B_2G_2 + C_2H_2 &= 0, \\ B_3G_3 + C_3H_3 &= 0, \\ B_4G_4 + C_4H_4 &= 0, \end{aligned}$$

the conditions in regard to the lines $2', 4', 5', 6'$ and $1, 3, 5, 6$ are formed by means of the diagram:

	1	3	5	6
$2'$	f_1	·	·	·
$4'$	$b_1g_2 - g_1b_4$	·	·	·
$5'$	·	·	·	·
$6'$	·	·	·	·

viz. we have the equations

first set,

$$\begin{aligned}f_1A_1 + g_1B_1 + h_1C_1 + a_1F_1 + b_1G_1 + c_1H_1 &= 0,\\ g_1B_1 + a_1F_1 + b_1G_1 + c_1H_1 &= 0,\\ g_1B_1 + b_1G_1 + c_1H_1 &= 0,\\ g_1B_1 + b_1G_1 + c_1H_1 &= 0;\end{aligned}$$

second set,

$$\begin{aligned}f_2A_1 + g_2B_1 + h_2C_1 + b_2G_1 + c_2H_1 &= 0,\\ g_2B_1 + c_2H_1 &= 0,\\ g_2B_1 + c_2H_1 &= 0,\\ g_2B_1 + c_2H_1 &= 0;\end{aligned}$$

third set,

$$\begin{aligned}f_3A_1 + g_3B_1 + h_3C_1 + b_3G_1 + c_3H_1 &= 0,\\ g_3B_1 + b_3G_1 + c_3H_1 &= 0,\\ g_3B_1 + b_3G_1 + c_3H_1 &= 0;\end{aligned}$$

fourth set,

$$\begin{aligned}f_4A_1 + g_4B_1 + h_4C_1 + a_4F_1 + b_4G_1 + c_4H_1 &= 0,\\ g_4B_1 + b_4G_1 + c_4H_1 &= 0,\\ g_4B_1 + b_4G_1 + c_4H_1 &= 0;\end{aligned}$$

and it is to be shown, that taking as given the coordinates of $2'$, $5'$, $6'$, that is $(A_1, B_1, C_1, G_1, H_1)$, $(B_3, C_3, F_3, G_3, H_3)$, $(B_4, C_4, F_4, G_4, H_4)$, we can find the coordinates of the remaining lines $4'$, 1, 3, 5, 6.

The first set of equations gives

$$g_1\,b_1 = b_1 \cdot g_1\,; \quad \begin{vmatrix} B_1 & C_1 & \cdot \\ B_3 & C_3 & H_3 \end{vmatrix};$$

viz. g_1, b_1, a_1 are proportional, but so only the ratios are material, they may be taken equal to the determinants $G_1H_3 - C_1H_3$, $B_3H_1 - B_1H_3$, $B_1C_3 - B_3C_1$, &c.; that is to signify these values respectively, the first equation gives f_1A_1, and the second equation gives a_1F_1; multiplying together these values, and writing $a_1f_1 = k_1g_1$, we find

$$(B_1B_3{\cdot}G_1G_3{\cdot}H_1H_3{\cdot}G_1H_3{\cdot}G_3H_1){\cdot}k_1(A_1,B_1,C_1,F_1,G_1,H_1)+B_3(B_1C_3{-}B_3C_1)F_2{\cdot}A_1+B_1(B_1C_3{-}B_3C_1)F_2 = 0.$$

Proceeding in a similar manner with the second set of equations, we have first

$$g_2,\ b_2,\ b_2 = \begin{vmatrix} B_1 & C_1 & G_1 \\ B_3 & C_3 & G_3 \end{vmatrix} \quad (\text{observe } k_2 = G_1B_3 - G_3B_1,\ = -c_1),$$

and then

$$(B_3B_1 \cdot G_1C_1 \cdot G_4C_4 \cdot G_4G_1 \cdot C_4C_1 \cdot G_1G_4 \cdot G_4G_1 + A_1F_1 \cdot B_3C_2 + B_1C_2 \cdot G_3 \cdot B_3G_1)k_2\,b_2 = 0.$$

The third set gives in like manner with the second set of equations, this becomes

$$g_3^2B_1B_3 + g_3b_3(B_1G_3 + B_3G_1 + A_1F_2) + b_3^2G_1G_3 = 0,$$

or since $g_3 \cdot b_3 = G_1 \cdot -B_1$, this is

$$G_1^2B_1B_3 - G_1B_1(B_1G_3 + B_3G_1 + A_1F_2) + B_1^2G_1G_3 = 0,$$

and similarly, the fourth set gives

$$g_4^2B_1B_3 + g_4b_4(B_1G_3 + B_3G_1 + A_1F_2) + b_4^2G_1G_3 = 0,$$

or since $g_4 \cdot b_4 = G_1 \cdot -B_1$, this is

$$G_1^2 B_1 B_3 - G_1 B_1 (B_1 G_3 + B_3 G_1 + A_1 F_2) + B_1^2 G_1 G_3 = 0,$$

and these last two results lead to the values of the ratios of $B_3 B_4$, $B_3 G_4 + B_4 G_3$, $G_3 G_4$ in the values containing the common factor $B_1 G_1$, viz. these are proportional to expressions containing the common factor $B_3 G_3 - B_4 G_4$, and taking them equal instead of merely proportional to the resulting expressions (which is allowable, since the absolute values are not material), we have

$$B_3 B_4,\ B_3 G_4 + B_4 G_3 + A_1 F_2,\ G_3 G_4 = B_1 B_3,\ B_1 G_3 + B_3 G_1,\ G_1 G_3,$$

$$(B_3 B_4,\ G_3 G_4,\ G_3 B_4 + B_3 G_4;\ p_4,\ q_4,\ r_4) = (B_1 B_3, G_1 G_3, B_1 G_3 + G_1 B_3; H_3,\ G_1,\ -H_1);$$

but the terms containing g_1, b_1 are $(B_1 g_1 + G_1 b_1)(B_4 g_2 + G_4 b_4), = -H_1 c_1 \cdot -H_4 c_4$, that is $H_1 H_4 c_1 c_4$; the whole equation is thus divisible by c_1, and omitting this factor, it becomes

$$g_1 (H_1 B_4 + H_4 B_1) + h_1 (G_1 H_4 + G_4 H_1) + a_1 (H_1 F_4 + H_4 F_1) = 0.$$

Proceeding in like manner with the result obtained from the second set of equations, this becomes

$$(B_1 B_3, C_2 C_4, C_1 G_1, C_1 G_4 + C_4 G_1, B_2 G_4 + B_4 G_2, B_4 G_4 + B_2 G_4 + B_2 C_1 (G_1, b_4, c_4)) = 0,$$

where the terms containing g_3, b_3 are $(B_2 g_3 + C_2 b_3)(B_4 g_2 + C_4 b_4)$, viz. the whole equation divides by h_3, and it then becomes

$$g_3 (B_3 C_4 + B_4 C_3) + h_3 (C_3 G_4 + C_4 G_3) + b_3 (G_3 B_4 + G_4 B_3) = 0.$$

Considering B_3, G_3, F_3 as given by the equations

$$B_3 B_4 = B_1 B_3, \quad G_3 G_4 = G_1 G_3, \quad B_3 G_4 + G_3 B_4 = A_1 F_3 = B_1 G_3 + B_3 G_1,$$

But we have

$$\begin{aligned} g_1 B_1 B_4 - b_1 H_4 H_1 H_4 &= B_4 H_1 (G_1 H_4 - G_4 H_1) - B_1 H_4 (G_1 H_4 - G_4 H_1), \\ &= B_4 H_4 (B_3 G_1 + G_3 B_1) - B_1 H_1 (B_3 G_4 + G_3 B_4) + C_1 B_3 = 0, \end{aligned}$$

and similarly

$$g_1 C_1 C_4 - b_1 G_4 G_1 G_4 = 0,$$

$$g_1 C_1 B_4 - b_1 H_4 G_1 G_4 = 0,$$

$$h_1 C_1 G_4 - g_1 B_4 H_1 G_4 = 0.$$

Moreover, writing $p_4 B_4 + q_4 G_4 + r_4 H_4$, we have

$$\begin{aligned} p_4 B_4 G_4 &= b_4 (G_4^2 - G_1 G_4), \\ q_4 G_4 G_1 &= b_4 (G_1 G_4 - G_4^2), \\ r_4 G_4 &= -b_4 (G_4 - G_1), \end{aligned}$$

and the few terms of the equation in question thus separately vanish; and the equation is consequently verified.

We may collect the results as follows:

Data are lines $1'$, 2, $3'$, 4; $5'$, $6'$;

and then, for the remaining lines 1, 3, 5, 6 at the common factor omitted in each case, the coordinates are as follows:

For $4'$,

$$B_1B_4 = B_1B_3, \quad G_3G_4 = G_1G_3, \quad B_3G_4 + G_3B_4 = A_1F_3 = B_1G_3 + B_3G_1,$$

$$A_1F_2 = B_1G_1 \cdot B_3G_3 - B_3G_1 \cdot B_1G_3, \quad A_1 = 0,$$

$$\begin{vmatrix} B_1 & C_1 & H_1 \\ B_3 & C_3 & H_3 \end{vmatrix}, \quad \begin{vmatrix} B_1 & C_1 & G_1 \\ B_3 & C_3 & G_3 \end{vmatrix}, \quad H_1H_3, \quad B_1B_3, \quad C_1C_3, \quad G_1G_3;$$

$$\begin{vmatrix} B_1 & C_1 & G_1 \\ B_3 & C_3 & G_3 \end{vmatrix} + \begin{vmatrix} C_1 & G_1 & H_1 \\ C_3 & G_3 & H_3 \end{vmatrix} \quad \Big(B_1G_3 + B_3G_1 = -2A_1F_2, \text{ identity}\Big).$$

For 1,

$$(g_1, b_1, a_1) = \begin{vmatrix} B_1 & C_1 & H_1 \\ B_3 & C_3 & H_3 \end{vmatrix}, \quad A_1F_1 = \begin{vmatrix} B_3 & C_3 & H_3 \\ B_1 & C_1 & G_1 \end{vmatrix},$$
$$(a_1f_1 + k_1g_1, \text{ identity}).$$

For 3,

$$(g_2, b_2, b_2) = \begin{vmatrix} B_1 & C_1 & G_1 \\ B_3 & C_3 & G_3 \end{vmatrix}, \quad A_1F_2 = \begin{vmatrix} B_3 & C_3 & G_3 \\ B_1 & C_1 & C_1 \end{vmatrix},$$
$$(a_2f_2 + k_2g_2, \text{ identity}).$$

For 5,

$$\begin{aligned} &g_3 = G_3, \quad b_3 = B_3, \quad A_1F_3 = B_1G_3 - B_3G_1, \\ &c_3 = G_3, \quad b_3 = 0, \qquad a_3 = \frac{A_1F_3G_3 + b_3(B_1G_1 + C_1H_1)}{B_3G_1 + b_1H_3}, \quad k_3 = 0. \end{aligned}$$

For 6,

$$\begin{aligned} &g_4 = G_4, \quad b_4 = -B_4, \quad A_1F_4 = B_1G_4 - B_4G_1, \\ &c_4 = 0, \qquad b_4 = 0, \qquad\quad a_4 = 0, \end{aligned}$$

and, for actual calculation, it is convenient to remark that as only the ratios are material, we may suppose without loss of generality that $c_1 = 1$, &c., respectively.

But results may be further reduced. Writing

$$(g_1, b_1, c_1) = \begin{vmatrix} B_3 & C_3 & H_3 \\ B_4 & C_4 & H_4 \end{vmatrix},$$

we have

$$-(g_1B_4 + b_1G_4 + h_1C_4) = C_1\left(\frac{B_3G_3}{B_1G_1}\right) + h_4C_4,$$

and many further simplifications follow, leading to compact final forms for C_1, H_3, etc.

[The remaining algebraic deductions on book pp. 321–325 *develop in the same manner the explicit formulas for the coordinates of* $2'$, $4'$, $5'$, $6'$ *and* 1, 3, 5, 6, *the closing display exhibiting the four quantities*

$$C_1 = \frac{C_1G_4\,p_4B_4 + b_4G_4 \cdot k_4 + B_4G_4}{B_4 \cdot p_4B_4 + h_4b_4 + k_4a_4},$$

together with the parallel expressions for G_1, F_1, H_1, H_3, *and the like for the alternative subscripts. The explicit verification is carried out on book p.* 325.*]*

I have thought it worth while to effect the numerical calculations for enabling the construction of a drawing or model. For this purpose taking X, Y, Z as ordinary rectangular coordinates, I write

$$\begin{aligned} x &= X + Y + Z - 10, \\ y &= Z, \\ z &= -X + Y + Z - 2, \\ w &= Y, \end{aligned}$$

that is, I take $1'$ and 2 to be lines in the plane of XY, defined by the equations $X+Y=10$ and $X-Y=-10$ respectively, and X and $3'$ to be lines in the plane of YZ defined by the equations $Z=10$, $Z+Z-2=12$ respectively. And I take $5'$ to be the line joining the points $(9,0,7)$ and $(-9,2,0)$; $4'$ the line joining the points $(9,0,7)$ and $(-2,8,0)$; $5'$ the line joining the points $(9,0,7)$ and $(-2,8,0)$; $6'$ the line joining the points $(9,0,7)$ and $(-2,8,0)$. We calculate then for each of the lines the upper coordinates of two points, and thence the six coordinates of the line; thus

	x	y	z	w	p	yw	A	B	C	F	G	H
$1'$	$0,0,-1,0$	$-10,0,0,1$	$0,0,0,0$	$0,0,0,0$			$0,$	$10,$	$50,$	$0,$	$0,$	-10
2	$0,0,-1,0$	$-10,0,0,1$	$0,0,0,0$				$0,$	$10,$	$50,$	$0,$	$0,$	-10
$3'$	$0,0,1,0$	$-10,0,0,1$					$0,$	$10,$	$50,$	$0,$	$0,$	-10
4	$0,0,1,0$	$-10,0,0,1$					$76,$	$30,$	$0,$	$0,$	$10,$	0

and, effecting the calculations for the remaining lines, we have

	A	B	C	F	G	H
$2'$	$\cdot$	24	$\frac{144}{15}$	$\cdot$	$\cdot$	$\frac{8}{15}$
1	$\frac{380}{15}$	60	35	$\frac{380}{15}$	0	3
3	127500	-720	0	$\frac{21}{15}$	140	-30
5	304	-54	6	0	7	0
6	$\frac{333}{5}$	-10	0	$\frac{12}{5}$	0	0

or reducing to integers, the values are

	A	B	C	F	G	H
$2'$	0	48	36	0	7	8
$4'$	76	26	2	0	0	1
1	1444	13452	6720	16443	-1131	0
3	463384	-2738	0	5	581	-114
5	9772	-552	0	21	132	0
6	5772	-3033	0	84	226	0

The line $(a,b,c;f,g,h)$ is given as the intersection of the two of the four planes

$$\begin{vmatrix} \cdot & h & -g & a \\ -h & \cdot & f & b \\ g & -f & \cdot & c \\ -a & -b & -c & \cdot \end{vmatrix} \begin{Bmatrix} X \\ Y \\ Z \\ 1 \end{Bmatrix} = 0,$$

or substituting for x, y, w the values $X+Y+Z-10$, Z, $-X+Y+Z-2$, Y, these become

$$\begin{vmatrix} \cdot & h & -g & a \\ -h & \cdot & f & b \\ g & -f & \cdot & c \\ -a & -b & -c & \cdot \end{vmatrix} \begin{Bmatrix} X+Y+Z-10 \\ Z \\ -X+Y+Z-2 \\ Y \end{Bmatrix} = 0,$$

viz., what is the same thing,

$$\begin{vmatrix} \cdot & 2y-a-c & 2y-f-h & \cdot & -2y \\ -2y+a+c & \cdot & a+b+c+f-h & \cdot & -c \\ -2y+f+h & -a-b-c-f+h & \cdot & \cdot & -f-h \\ 2y & c+c & f+h & \cdot & \end{vmatrix} \begin{Bmatrix} X \\ Y \\ Z \\ 1 \end{Bmatrix} = 0.$$

And substituting, we have the equations of the several lines, viz.

$$
\begin{array}{ll}
(1') & X+Y=10, \quad Z=0; \\
(2) & -X+Y=10, \quad Z=0; \\
(3') & -X+Y=2, \quad Z=10; \\
(4) & X+Y=10, \quad Z=0;
\end{array}
$$

$$
(2') \quad \begin{vmatrix} \cdot & \cdot & 49 & 5 \\ -49 & \cdot & \cdot & -36 \\ -5 & 36 & \cdot & \cdot \end{vmatrix} \begin{Bmatrix} X \\ Y \\ Z \\ 1 \end{Bmatrix} = 0;
$$

$$
(4') \quad \begin{vmatrix} \cdot & -55 & 26 & -7 \\ -55 & \cdot & \cdot & -3 \\ 26 & \cdot & \cdot & -1 \\ 7 & 26 & -3 & \cdot \end{vmatrix} \begin{Bmatrix} X \\ Y \\ Z \\ 1 \end{Bmatrix} = 0;
$$

and so on for $(5')$, $(6')$, (1), (3), (5), (6) with the corresponding numerical matrices given on book p. 328.

[Page 329 *of the original reproduces the numerical coordinates of the various intersection points* $11', 12', 13', \ldots,$ *in fractional form (numerators and denominators displayed on separate lines), and adds:]*

viz., the coordinates of $12'$ (intersection of lines 1 and $2'$) are $\left(\frac{611}{132}, \frac{-554}{132}, \frac{-216}{132}\right)$, and so in other cases; where there is no divisor the coordinates are integers. I find however, on laying down the figure, that the lines 3 and 4, $5'$ and 6 come so close together, that the figure cannot be obtained with any accuracy.

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[460] Note on Mr Frost's paper on the direction of lines of curvature in the neighbourhood of an umbilicus.

*[From the Quarterly Journal of Pure and Applied Mathematics, vol. X. (*1870*), pp.* 111–113.*]*

I remark as follows:

1. In regard to a quartic surface, it is not, I think, correct to say that the generation through an umbilic (= curve of curvature) notwithstanding that, in fact in the present case there will generate 5 in one plane and consequently intersect. The way in which these generatrices as quasi-curves-of-curvature present themselves is as follows:

The curve of curvature satisfy a certain differential equation, the complete integral of which gives those curves as the intersections of the given quadric surface by the series of confocal quadrics

$$\frac{x^2}{a^2-h} + \frac{y^2}{b^2-h} + \frac{z^2}{c^2-h} = 1,$$

h being the constant of integration. The singular solution of the differential equation, or envelope of the series of curvatures denoted as above, gives the umbilical generatrices.

2. In regard to a surface in general, I think it must be considered, not that there pass through the umbilic three distinct curves, but that there pass through the umbilic a single point, or rather a point at which there are in general three distinct directions of the curve. The umbilicar curve of curvature is in fact present itself as the curve belonging to a certain value of the constant of integration h, in order that the curve of curvature may pass through a given point on the

surface; a most simple example will, in order that there are in general distinct directions of the curve.

But, as an umbilic is a point for which (as in effect shown, p. 81, for the particular case of a quadric surface) the two values of h become equal; that is, through the umbilic only a singular curve of curvature; but $\dfrac{dy}{dx}$ is determined by a cubic equation, and the umbilic is thus (as just mentioned) a point at which there are in general three distinct directions of the curve.

3. Some researches on the subject are contained in my paper "On Differential Equations and Umbilici," *Phil. Mag.*, vol. XXVI. (1863), pp. 373–379 and 441–452, [230]. It is noticeable that in the integral equations which I have there obtained for the differential equation $dy(p^3-1)+(a-c)pq=0$, and the more general form $(\mathrm{d}e+p)(p+q)=0$, and the umbilic itself, the differential equation breaks up into three distinct curves; and the same is the case with the cubic equation, that the umbilic does not occur in the surface $xyz=1$ presently referred to.

4. In the paper "Mémoire sur les surfaces orthogonales," *Liouv.*, t. XII. (1847), pp. 241–254, M. Serret has given two very remarkable cases of three systems of surfaces intersecting each other at right angles, and consequently in the curves of curvature of each system. It was only on referring to this paper, in connexion with that of Mr Frost, that I perceived an obvious enough simplification of M. Serret's formulae, whereby it appears that the curves of curvature on the surface $xyz=1$ are given as the intersection of this surface with the series of surfaces

$$h=(x^2+xy^2+xz^2)^{\frac{1}{3}}+(x^2+xy^2+xz^2)^{\frac{1}{3}},$$

where ω is an imaginary cube root of unity; the rationalised equation is of the twelfth order in (x,y,z), and for the particular value $h=0$ is $(y^3-x)^3+(x^3-y^2)^3+(xy^3-1)^3$. The point $x=y=z=1$ is obviously an umbilic on the surface $xyz=1$, and the corresponding value of h being $h=0$, the equation just obtained determines the umbilicar curves of curvature, viz., combining therewith the equation $xyz=1$, we infer that the surface have here hyperbolic curves

$$(y=z,\ xy^2=1),\quad (z=x,\ yz^2=1),\quad (x=y,\ zx^2=1).$$

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[461] On the Geometrical Interpretation of the Covariants of a Binary Cubic.

*[From the Quarterly Journal of Pure and Applied Mathematics, vol. X. (*1870*), pp.* 148–149.*]*

Consider the binary cubic $U=(a,b,c,d\,|\,x,y)^3$, and the discriminantal (invariant)

$$\nabla=a^2d^2-6abcd+4ac^3+4b^3d-3b^2c^2,$$

the Hessian

$$H=(ac-b^2)x^2+(ad-bc)xy+(bd-c^2)y^2,$$

and the cubicovariant

$$\Phi=\begin{aligned}&(\ \ a^2d-3abc+2b^3)x^3\\&+(3abd+6ac^2-9b^2c)x^2y\\&-(3acd+6b^2d-9bc^2)xy^2\\&-(a^2d-3acd+2c^3)y^3,\end{aligned}$$

connected by the identical relation

$$\Phi^2=-\nabla U^2-4H^3.$$

Then if we regard (a, b, c, d) as the coordinates of a point in space, but (x, y) as variable parameters, the equation

$$\nabla = 0$$

represents a quartic torse, having for its cuspidal curve the plane cubic $ad - bc = 0$, $bd - c^2 = 0$, the equation

$$U = 0$$

is that of the tangent plane to the torse along the line $ad + 2bc + cy = 0$, $bd^2 + 2acy + dy^2 = 0$; the line is the cuspidal curve in the above considered as line coordinates; viz. $a : b : c : d = ay^3 : -y^2 : y : -1$, the equation

$$H = 0$$

is that of a quadric surface having the first point of the osses point in regard to the torse.

The equation $\Phi^2 = \nabla U^2 - 4H^3$, gives $\Phi = -4H^3$, a result which implies that $\Phi = 0$ is a certain torse required twice, and that $U = 0$, $\Phi = 0$ is the same equator repeated twice. The curve in question is at any rate on the surface $\Phi = 0$; and conversely, the curve is on the surface $\Phi = 0$ regarded as the envelope of the plane $U = 0$, viz. in $(2, 3)$-plane, &c., what is the same thing.

The equation $\Phi^2 + \nabla U^2 + 4H^3 = 0$, gives $\Phi^2 = -4H^3$, a result which implies that $\Phi = 0$ is a certain torse repeated twice, that is, $H = 0$, $\Phi = 0$, the whole equation in this divisible by c, and omitting this factor, it becomes

$$g(H_1B_4 + H_4B_1) + h(G_1H_4 + G_4H_1) + a(H_1F_4 + H_4F_1) = 0.$$

Proceeding in like manner with the result obtained from the second set of equations, this becomes

$$(B_1B_3, C_2C_4, C_1G_1, C_1G_4 + C_4G_1, B_2G_4 + B_4G_2, B_4G_4 + B_2G_4) = 0.$$

[End of paper 461; the manipulations on book p. 333 verify the geometrical interpretation by reducing to the form $MU = (a^2, b, c, d \mid x, y)^3 \cdot M$, $\Phi = U$ and complete the discussion of the cubicovariant.]

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[462] A Ninth Memoir on Quantics.

[From the Philosophical Transactions of the Royal Society of London, vol. CLXI. (for the year 1871), pp. 17–50. Received April 7. —Read May 19, 1870.]

It was shown not long ago by Professor Gordan that the number of the irreducible covariants of a binary quantic of any order is finite (see his memoir "Beweis dass jede Covariante und Invariante einer binären Form eine ganze Function mit numerischen Coefficienten einer endlichen Anzahl solcher Formen ist," *Crelle*, t. LXIX. (1869), Memoir dated 8 June 1868), and in particular that for a binary quintic the number of irreducible covariants (including the quintic and the invariants) is = 23, and that for a binary sextic the number is = 26. From the theory given in my "Second Memoir on Quantics," *Phil. Trans.*, 1856, [141], I derived the conclusion, which, as it now appears, was erroneous, that for a binary quintic the number of irreducible covariants was infinite. The theory requires, in fact, a modification, by reason that certain linear relations, which I had assumed to be independent, are really not independent, but, on the contrary, linearly connected together; the interconnexion in question does not occur in regard to the quadric, cubic, or quartic; and for these cases respectively the theory is as it stands; for the quintic the interconnexion first presents itself in regard to the degree 8 in the coefficients and order 14 in the variables, viz. the theory gives correctly the number of covariants of any degree not exceeding 7, and also those of the degree 8 and order less than 14; but for the order 14 the theory as it stands gives a non-existent irreducible covariant $(a, \dots \mid x, y)^{14}$, viz. of the form in question $10 - 6$, $= 4$ asyzygetic covariants; but the number of asyzygetic covariants being $= 5$, there is left, according to the theory, 1 irreducible covariant of the form in question. The fact is that in applying the theory, the composite covariants are equivalent to 5 independent syzygies only, the composite covariants are equivalent to $10 - 5$, $= 5$, the full number of the asyzygetic covariants. And similarly the theory as it stands gives a non-existent of the asyzygetic covariants.

[Article No. 326 *continues on book pp.* 334–335 *with Cayley's revised treatment of the issue; thence Articles* 327–365 *develop, on book pp.* 335–353*, the systematic proof. The articles fall into the following sections.]*

Article Nos. 328 **to** 332**.** ***Reproduction of my original Theory as to the Number of the Irreducible Covariants.***

328. I reproduce to some extent the considerations by which, in my Second Memoir on Quantics, I endeavoured to obtain the number of the irreducible covariants of a given binary quantic $(a, b, \dots | x, y)^n$.

Considering in the first instance the covariants as functions of the coefficients $(a, b, c, \dots)$ without regarding the variables (x, y), and attending only to the following properties — 1°, without regarding the variables[1], a covariant is a rational and integral homogeneous function of the coefficients $(a, b, c, \dots)$ of the properties — 1°, a covariant is a rational and integral homogeneous function of the coefficients $(a, b, c, \dots)$; and 2°, the coefficients of a given degree μ are linearly independent of the coefficients of a less degree; and that a covariant not connected by linear relations as above; and it is assumed that we know the number of the *irreducible* covariants; and it is in the same manner that we know the number of the asyzygetic covariants in regard to a given quantic, and that as the degree μ is increased successively to 1, 2, 3, ... there is a certain function P, Q, R, ... for each degree, and that a function not so obtained must be an *irreducible* covariant; and it is assumed that we know the number of the asyzygetic covariants of the degree μ as a given function P_μ of the coefficients a_i, ...

[The text continues through Articles 329–346 *on book pp.* 336–345*, treating successively: the binary cubic with its three syzygies; the binary quartic with its three covariants and one syzygy; the quintic with its* 23 *irreducible covariants (Table No.* 87*, identifying the* 23 *in terms of Gordan's notation* A*,* B*,* C*, ...,* W*); Table No.* 88*, giving the actual generation, by successive transvectants, of each of the* 23 *forms (degrees* 1 *through* 8*); the related Tables Nos.* 89 *and* 90 *exhibiting the relations and complete linear expressions of the syzygies among the products of the covariants. The reasoning is throughout that of Gordan's Beweis but cast in the notation* $(f, f), (f, (f, f)), \dots$ *of the Second Memoir.]*

Article Nos. 347 **to** 365**.** ***Sketch of Professor Gordan's proof for the finite Number,*** $= 23$***, of the Covariants of a Binary Quintic.***

347. I propose to reproduce the leading points of Professor Gordan's proof that the binary quintic $(a, b, c, d, e, f \mid x, y)^5$ has a finite system of 23 covariants, viz. a system such that every other covariant whatever is a rational and integral function of these 23 covariants.

348. DERIVATION. Consider for a moment any two binary quantics ϕ, ψ of the same or different orders, and which may be either independent or, as will be the case here, identical or connected

[1]For this case of covariants, α_1 is of course $= 1$; but in the investigation the term covariant stands for any function satisfying the conditions 1° and 2°.

by some relation f. We may form the derivatives

$$\begin{aligned}
&(\phi,\psi) = \phi\psi,\\
&(\phi,\psi)' = 1{\cdot}2\ \phi\psi,\ \partial_x\phi{\cdot}\partial_y\psi,\ -\partial_y\phi{\cdot}\partial_x\psi,\\
&(\phi,\psi)'' = 1{\cdot}2\ \phi\psi,\ \partial_x\phi,\ \partial_y\phi{\cdot}\partial_x\psi,\ -2\partial_x\partial_y\phi{\cdot}\partial_x\partial_y\psi + \partial_y^2\phi{\cdot}\partial_x^2\psi,
\end{aligned}$$

where, however, there is no occasion to use the notation $(\phi,\psi)'$ (as this is simply the product $\phi\psi$), and the succeeding derivatives may (when there is no risk of ambiguity) be written more shortly $(\phi\psi)$, $(\phi\psi)'$, &c.; in all that follows the term "derivative" (Gordan's *Ueberschiebung*) is to be understood in this special sense.

349. The degree of the derivative $(\phi\psi)^r$ is the sum of the degrees of ϕ, ψ; the order of the derivative is the sum of the orders of the constituents ϕ, ψ; the order of the derivative is less than this by $2r$; it being understood throughout that the next degree refers to the coefficients, and the order to the variables.

350. Theorem A. The covariants of a quantic f of a given degree m can be all of them obtained by derivation from f and the covariants of the next inferior degree $(m-1)$.

In particular for the degree 1 the only covariant is the quantic f itself; the degree 2 the covariants of the third degree consist of the quantic f itself in succession, the covariants of the third degree are each of them obtained, by derivation from f and the covariants of the second degree.

351. Suppose that the covariants of the second degree $(f,f)^r$, $(f,f,f)^r$, $(f,(f,f))^r$, … are in this order represented by $\beta_1,\beta_2,\beta_3,\ldots$, then the covariants of the third degree written in the order

$$(\beta_1 f)^r,\ (\beta_1 f)^{r'},\ \ldots,\ (\beta_2 f)^r,\ (\beta_2 f)^{r'},\ \ldots,\ (\beta_3 f)^r,\ (\beta_3 f)^{r'}\ldots$$

may be represented by $\delta_1,\delta_2,\ldots$, and the covariants of the fourth degree written in the order

$$(\gamma_1 f)^r,\ (\gamma_1 f)^{r'},\ \ldots,\ (\gamma_2 f)^r,\ (\gamma_2 f)^{r'},\ \ldots,\ (\gamma_3 f)^r,\ (\gamma_3 f)^{r'},\ldots$$

may be represented by $\delta_1,\delta_2,\ldots$, and so on.

Observe that each term μ_r is a derivative $(\lambda f)^r$, the derivatives of λ with f being assumed to be an earlier term; the derivative with a less index of derivation is earlier than that with a greater index of derivation, or, what is the same thing, those, those are earlier which have a less index r, and which have a greater index of the higher order.

352. The series $\mu_1,\mu_2,\mu_3,\ldots$ is not asyzygetic; we make this so by considering in succession whether the several terms $\mu_1,\mu_2,\ldots$ respectively are expressible as linear functions of the other terms, or omitting every term which is so expressible. The reduced series obtained is a series of terms which is in fact a complete asyzygetic series of covariants of the degrees in question.

353. The quadricovariants $(a,b,c\,|\,x,y)^2$ are proportional, but as only the ratios are material, they may be taken equal to the determinants $G_1H_3-G_3H_1$, $B_3G_1-B_1G_3$, $B_1C_3-B_3C_1$, &c. They form a system T of T-series, and the second equation gives $\alpha_1 f_1$; multiplying together these values, and writing $a_1f_1 = k_1g_1$, we find

$$(B_1B_3\cdot G_1G_3\cdot H_1H_3\cdot G_1H_3\cdot G_3H_1)\cdot k_1(A_1,B_1,C_1,F_1,G_1,H_1)=0.$$

354. For the cubic $(a,b,c,d\,|\,x,y)^3$ the number of asyzygetic covariants $a^\mu x^\nu$ is

$$\text{coeff. } a^\mu x^\nu \text{ in } \frac{1-\frac{1}{x^2}}{(1-a)(1-ax)\left(1-\frac{a}{x}\right)},$$

or transforming as before, this is

$$\text{coeff. } a^\mu x^\nu \text{ in } \frac{1-x^2}{(1-ax)(1-ax^3)(1-a^3)}.$$

The development is

$$\begin{array}{l|l}
1 & \\
+\,ax^3 & -\tfrac{1}{x^2} \\
+\,a^2(x^2+1) & +\,a\cdot\left(\tfrac{1}{x}\right) \\
+\,a^3(x^3+x+1) & +\,a^2\left(\tfrac{1}{x}+1\right) \\
+\,a^4(x^2+x^4+1) & +\,a^3\left(\tfrac{1}{x}+\tfrac{1}{x^3}\right) \\
\vdots & +\,a^4\left(\tfrac{1}{x}+\tfrac{1}{x^3}+1\right)
\end{array}$$

which is

$$= A(x) - \tfrac{1}{x^2}\,A\left(\tfrac{1}{x}\right),$$

where

$$A(x) = \frac{1}{(1-ax^3)(1-ax)},$$

and, since $\tfrac{1}{x^2}\,A\left(\tfrac{1}{x}\right)$ contains only negative powers, the required number is

$$= \text{coeff. } a^\mu x^\nu \text{ in } \frac{1}{(1-ax^3)(1-ax)},$$

indicating that the covariants are powers and products of (ax^3 and ax), the quadric itself, and the discriminant. Compare Second Memoir, No. 40, according to which, writing therein a for a, the number of asyzygetic covariants is

$$= \text{coeff. } a^\mu x^\nu \text{ in } \frac{1}{(1-ax)(1-x)}.$$

[Article numbers 355 *onwards (New formula for the number of Asyzygetic Covariants) continue on book pp.* 338–345 *in the same vein, expressing the number of asyzygetic covariants of the quadric, cubic, quartic and quintic as the coefficient of* $a^\mu x^\nu$ *in a generating fraction with a quadric or quartic Hessian-like denominator; these articles culminate in the explicit verification (book p.* 344*) that for the quintic the number is finite and* = 23*, and in the introduction (book p.* 345*) of* ***Article No.* 337*. Identification of the* 23 *Fundamental Covariants****, with Table No.* 87 *relating each of the* 23 *irreducible covariants to its Gordan symbol A, B, C, …, W.]*

[Tables Nos. 87, 88 *and the supplementary Table No.* 89 *(giving the syzygies and interconnexions of the syzygies among the products of the* 23 *covariants) occupy book pp.* 341–346*. They are intricate purely typographical exhibits whose intelligent reproduction is deferred to a later chunk; the present chunk closes with the opening of Articles* 347–365*, "Sketch of Professor Gordan's proof for the finite Number,* = 23*, of the Covariants of a Binary Quintic," as begun above.]*

Article No. 363. For U equal to any one of the last eleven values, the form is $(Q, S)R$ which is $= R(QS) + S(QR)$, and is thus a function of covariants of a lower degree; there remain the derivatives formed with two of the functions f, ϕ, i, j, p, τ, or of one of these with α or γ. But these are all U other than the derivatives

$$(fj),\ (\phi j),\ (\phi p),\ (\phi\tau),\ (p\tau);\ (f\alpha),\ (\phi\alpha),\ (j\alpha),\ (p\alpha);\ (f\gamma),\ (\phi\gamma),\ (j\gamma),\ (p\gamma),\ (\tau\gamma),$$

and since $\gamma = (\tau\alpha)$, the derivatives containing γ will depend upon covariants of a lower degree; there remain therefore only (fj), (ϕj), (ϕp), $(\phi\tau)$, $(p\tau)$; $(f\alpha)$, $(\phi\alpha)$, $(j\alpha)$, $(p\alpha)$: each of these can be actually calculated in the form $F(U)$.

Hence finally, assuming that every covariant of a degree inferior to m is $F(U)$, it follows that every covariant of the degree m is $F(U)$; whence every covariant whatever is $F(U)$, viz. it is a rational and integral function of the 23 covariants U.

364. It will be observed that, writing A, B, C for P, Q, i, the proof depends on the theorems

$$\begin{array}{ll} ((AB),\ C), & \text{a linear function of } A(BC)',\ B(CA)',\ C(AB)', \\ (AB,\ C)' & \text{do. do. do.} \\ ((AB),\ C)' & \text{do. do. } ((AC)', B),\ ((BC)', A),\ B(AC)',\ C(AB)', \end{array}$$

which are theorems relating to any three functions A, B, C whatever.

365. I remark upon the proof that the really fundamental theorem seems to be that which I have called theorem A. As to the forms W it is difficult to see *à priori* why such forms are to be considered, or what the essential property involved in their definition is; and in fact in a more recent paper, "Die simultanen Systeme binären Formen" (*Math. Annalen*, t. II. (1869), see p. 256), Professor Gordan has modified the definition of the forms W by omitting the condition that the order of the function shall exceed n; if it were possible further to omit the condition of at least one index being $=$ or $< \frac{1}{2}n$, and so only retain the conditions $-n-i$, $n-j$, &c., each of them > 0, then the essential property of the forms W would be that any such form was a rational and integral function of the special covariants formed, as above, by means of the quantic of the next inferior order. And moreover, as regards the theorem B, there seems something indirect and artificial in the employment of such a property; one sees no reason why, when a system of irreducible covariants is once written down, it should not be possible to show that the derivatives of $F(U)$ with the original quantic are each of them $F(U)$, instead of having to show this in regard to the derivatives of $F(U)$ with the several covariants χ: as regards the quintic, where there is a single covariant χ, the quadric function i, there is obviously a great abbreviation in this employment of i in place of f; but for the higher orders, assuming that the proof could be conducted by means of the quantic itself, it does not appear that there would be even an abbreviation in the employment in its stead of the several covariants χ. The like remarks apply to the proof in the last-mentioned paper. I cannot but hope that a more simple proof of Professor Gordan's theorem will be obtained — a theorem the importance of which, in reference to the whole theory of forms, it is impossible to estimate too highly.

(End of chunk: PDF pages 326–375, book pages 304–353.
Paper 457 in full,
paper 458 in full,
paper 459 in full,
paper 460 in full,
paper 461 in full,
paper 462 opening (Articles 326–365, book pp. 334–353).)